AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL **Submission:** model_6

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{7}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

A complete, correct solution. It establishes the common residue modulo g, shows $c^{k-1} \equiv -1 \pmod{g}$ with c a unit, proves $g < 500000$ for $k = 4$, excludes $g \geq 998286$ for $k = 3$, and provides a construction for 998285. Only trivial omissions remain (e.g. not spelling out why r odd gives $2 \mid r^2 + 1$ in the $m = 2$ case), which do not affect the mark. Scores **7**.